

MINOR PROJECT REPORT

WEATHER MONITORING AND ENVIRONMENTAL SAFETY STATION

Arduino Uno Based Tinkercad Simulation

**Submitted as a Minor Project under the IoT & Robotics Training Program
July Batch 2026**

Prepared from the implemented Tinkercad circuit, component list, and project guidelines.

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1. Abstract

This project presents the design and simulation of an Arduino Uno based environmental monitoring and safety station. The implemented Tinkercad circuit combines an Arduino Uno R3, 16x2 LCD, TMP36 temperature sensor, photoresistor (LDR), gas sensor, potentiometer, resistors, and a piezo buzzer. The system is intended to observe environmental conditions in real time, present sensor information to the user, and provide an audible warning when a monitored condition crosses a programmed threshold. The design therefore demonstrates the core embedded-system cycle of sensing, processing, displaying, and alerting.

The project was developed as a simulation rather than a fabricated physical prototype. Tinkercad was used to verify the circuit arrangement and visualize the connections. The current implementation measures temperature, light intensity, and gas-related sensor output. The project guidelines for the Weather Monitoring Station require temperature and humidity as mandatory measurements; the present simulation does not show a dedicated humidity sensor, so humidity is identified in this report as a current limitation and a necessary enhancement for full compliance with that requirement. (Project Guidelines, Section 2: Weather Monitoring Station)

2. Introduction

Environmental parameters influence comfort, safety, and the operation of electronic and industrial systems. A small embedded monitoring station can provide a practical way to collect these measurements continuously and communicate the results to a user. In this project, the Arduino Uno acts as the central controller. Analog sensor outputs are sampled by the microcontroller, processed in software, and presented through a 16x2 LCD. A piezo buzzer is included as an alert actuator for abnormal conditions.

The selected architecture is intentionally simple and educational. It demonstrates sensor interfacing, analog data acquisition, threshold-based decision making, LCD interfacing, actuator control, and simulation-based testing. These are foundational skills for IoT and robotics projects and can later be extended with cloud connectivity, data logging, dashboards, or wireless communication.

3. Problem Statement

A basic monitoring system should be able to observe environmental conditions continuously and provide information in a form that can be understood immediately. Manual observation is intermittent and may not provide an alert when conditions become unsafe. The proposed solution addresses this problem by combining multiple sensors with an Arduino-based controller and a local display/alarm interface.

4. Project Objectives

- Acquire environmental readings using the TMP36, LDR, and gas sensor in a continuous control loop.
- Display the measured information on a 16x2 LCD for local monitoring.
- Use a piezo buzzer to provide an audible indication when programmed threshold conditions are met.
- Demonstrate safe and organized interconnection of sensors, display, controller, and output devices in Tinkercad.
- Provide a foundation for future expansion toward a complete weather station with humidity measurement and IoT connectivity.

5. System Overview and Working Principle

The system follows a sensing-processing-output architecture. Sensors form the input layer, the Arduino Uno forms the processing layer, and the LCD/buzzer form the output layer. The Arduino repeatedly samples the connected analog inputs, performs the required conversion or comparison, updates the display, and controls the alarm output according to programmed logic.

Block	Main Element	Role
Input layer	TMP36 temperature sensor	Provides an analog signal related to temperature.
Input layer	Photoresistor (LDR)	Provides an analog signal representing relative ambient light.
Input layer	Gas sensor	Provides an analog gas-related signal for threshold monitoring.
Processing layer	Arduino Uno R3	Reads sensors, processes readings, updates outputs, and executes control logic.
Output layer	16x2 LCD	Displays sensor information and status messages to the user.
Output layer	Piezo buzzer	Provides an audible warning during an alert condition.
Adjustment/support	Potentiometer and resistors	Support LCD operation/contrast and signal/interface connections.

6. Component List

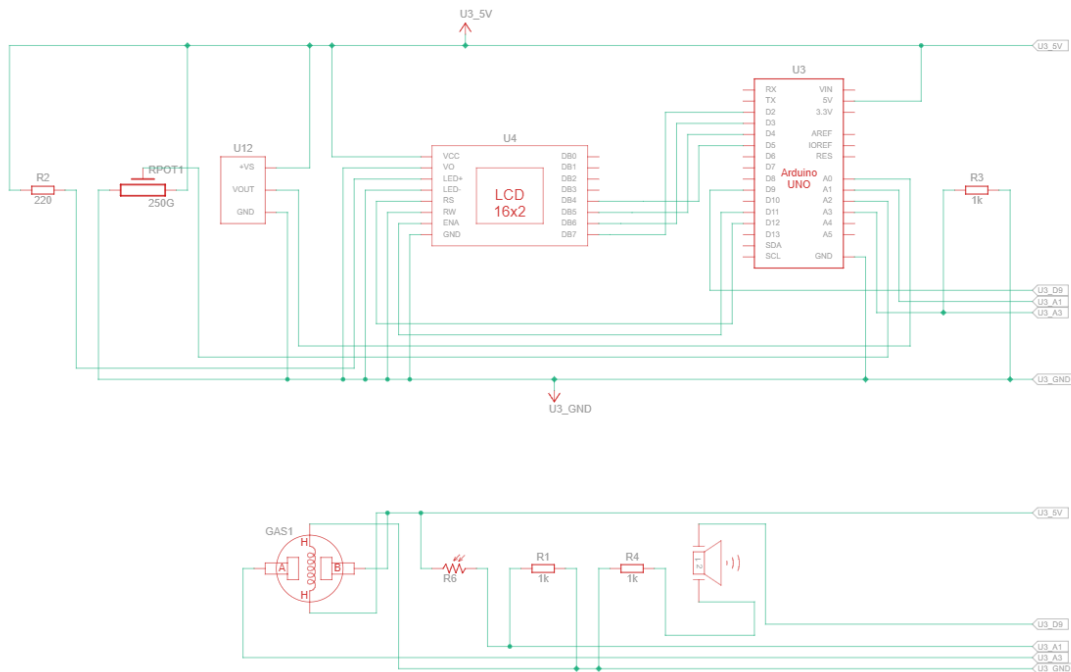


Figure 2. Component list shown in the project design. The report preserves the component names used in the provided simulation/BOM.

The component list in the supplied design identifies Arduino Uno R3, Breadboard Small, LCD 16x2, a 220 ohm resistor, TMP36 temperature sensor, three 1 kOhm resistors, a potentiometer, gas sensor, photoresistor, and piezo. The visible BOM should be treated as the reference for the implemented simulation; individual component values should be rechecked before physical construction because the displayed potentiometer notation is unclear in the source image.

7. Circuit Diagram and Circuit Explanation

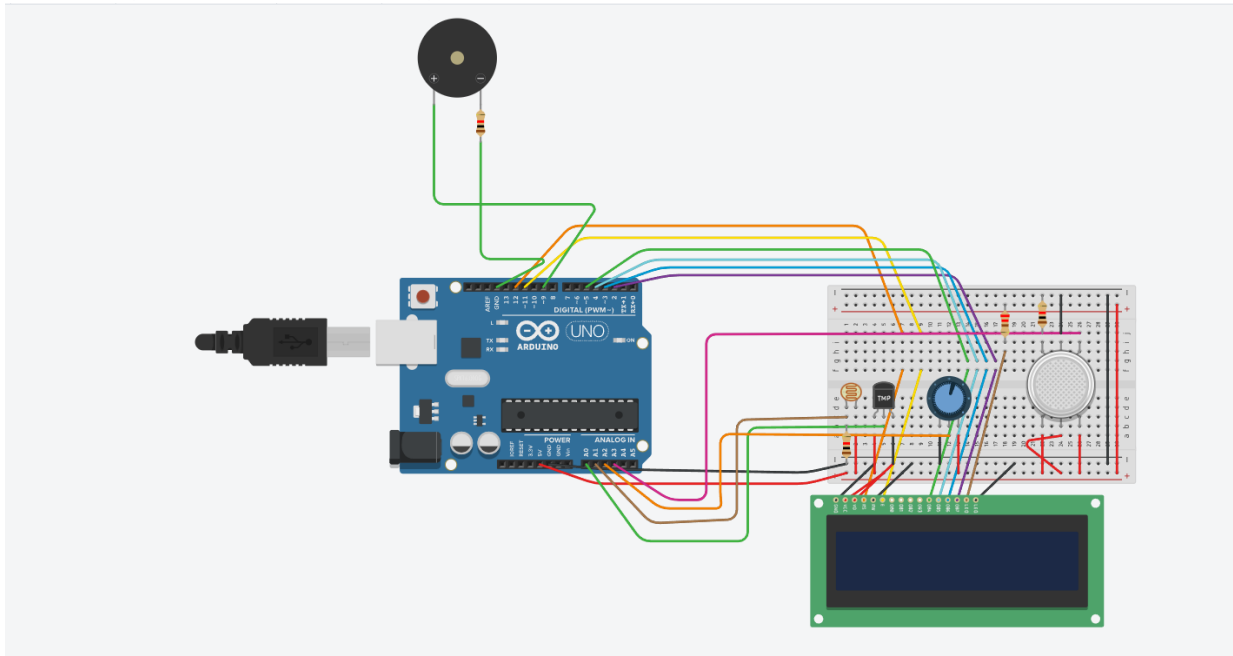


Figure 3. Tinkercad schematic/circuit view used as the basis for the project implementation.

7.1 Power Distribution

The Arduino Uno provides the regulated supply and ground references used by the sensor and interface network in the simulation. The schematic shows common 5 V and ground rails. Shared power and ground references are essential because the analog sensor voltages are interpreted relative to the Arduino ground.

7.2 Temperature Sensing

The TMP36 temperature sensor is connected as an analog input device. Its output voltage changes with temperature. The Arduino samples this analog voltage through its ADC and the program can convert the reading into a temperature value for display or threshold comparison.

7.3 Light Sensing

The photoresistor forms a light-dependent sensing element. Its resistance varies with incident light, and the resulting voltage at the Arduino analog input changes accordingly. The project therefore provides a relative indication of ambient brightness that can be used for environmental observation or as a future automation input.

7.4 Gas Detection and Alarm

The gas sensor supplies an analog signal to the controller. Because the simulation is threshold based, the program can compare the sensed value with a predefined limit. When the value is interpreted as unsafe, the Arduino drives the piezo buzzer to generate an audible alert. The exact safe/unsafe threshold should be calibrated against the chosen sensor and application before real-world deployment.

7.5 LCD Interface

The 16x2 LCD is connected to the Arduino for local display. A potentiometer is included to adjust LCD contrast. The design uses the LCD as the human-machine interface so that the user can observe measurements without opening the program or relying solely on the simulation workspace.

8. Hardware Design

Although the project is currently implemented in Tinkercad, the circuit has been organized with physical hardware implementation in mind. The Arduino Uno is positioned as the controller, the breadboard provides a common interconnection platform, and the display/sensors are arranged around the controller. The design emphasizes short signal paths, shared ground, and readable separation between sensing and output devices.

Subsystem	Design Consideration	Expected Result
Controller	Arduino Uno R3 with common 5 V and GND references	Stable coordination of all sensing and output operations.
Temperature	TMP36 analog interface	Continuous temperature observation.
Light	LDR/resistive sensing circuit	Relative ambient-light measurement.
Gas	Analog gas sensor input	Gas-related level monitoring and threshold alert.
Display	16x2 LCD with contrast potentiometer	Immediate local readout.
Alarm	Piezo buzzer controlled by Arduino output	Audible warning during an alert condition.
Interconnect	Breadboard and jumper wires	Flexible prototyping and simulation of wiring.

9. Software Implementation

The software follows a repetitive monitoring loop. At startup, the Arduino configures the LCD and input/output pins and prepares the system for sensing. The loop then reads the connected sensor channels, processes the raw values, updates the LCD, evaluates alarm conditions, and waits briefly before repeating. This modular structure is appropriate for a small embedded monitoring application because it is easy to test and extend.

1. Initialization: configure sensor inputs, buzzer output, and LCD interface.
2. Sampling: read the analog values from temperature, light, and gas-related channels.
3. Conversion/processing: convert raw ADC values into useful engineering or relative values as required by the chosen sensor.
4. Display update: show the latest readings and system status on the LCD.
5. Decision: compare monitored values with programmed thresholds.
6. Alert: activate the piezo buzzer when an unsafe condition is detected; otherwise keep it inactive.
7. Loop: repeat the process continuously so that the information remains current.

10. System Workflow / Control Algorithm

The following algorithm represents the implemented concept and can be mapped directly into an Arduino sketch using `setup()` and `loop()` functions.

8. BEGIN
9. Initialize Arduino pins and the 16x2 LCD.
10. Initialize/prepare temperature, light, and gas sensor inputs.
11. Read temperature sensor value.
12. Read LDR value.
13. Read gas sensor value.
14. Convert/process readings as required.
15. Display readings and status on LCD.
16. Compare gas/safety values against programmed threshold(s).
17. If threshold is exceeded, turn buzzer ON; otherwise turn buzzer OFF.
18. Wait for the next sampling cycle.
19. Repeat continuously.

11. Testing Methodology

Testing was considered from the perspective of individual sensor response, output response, and complete-system behavior. The Tinkercad environment makes it possible to vary simulated conditions without risking physical components. The required project guidelines also emphasize testing the mandatory functions, verifying sensor readings, checking hardware connections, testing relevant operating conditions, and checking unexpected readings/edge cases. (Project Guidelines, Section 8: Testing Requirements)

Test Case	Stimulus/Condition	Expected Behaviour	Observed/Design Result
T1	Normal temperature input	Temperature value is sampled and shown.	System provides continuous temperature monitoring.
T2	Change in light level	LDR reading changes and is available to software/display.	Light-level input responds to simulated illumination changes.
T3	Increase gas sensor input	Gas-related value increases and threshold logic evaluates it.	Alarm can be triggered when configured limit is reached.
T4	Alert condition	Buzzer output becomes active.	Audible warning function is included in the design.
T5	LCD operation	Display remains readable and updates with current values.	LCD is used as the primary local interface.
T6	Restart/power-up	System initializes without manual intervention.	Arduino startup logic reinitializes sensors/display.

12. Results and Discussion

The Tinkercad simulation demonstrates the intended integration of the sensors, Arduino Uno, LCD, and buzzer in one monitoring station. The circuit view shows a coherent electrical architecture with common power/ground rails and multiple sensor/output connections. The design is suitable as an educational prototype because the system behavior can be changed and observed without physical component damage.

A key result is the combination of monitoring and alerting in a single system. The LCD provides direct feedback, while the buzzer provides immediate attention when the programmed condition becomes abnormal. The use of the LDR and gas sensor also broadens the system beyond a temperature-only station, although these additions do not replace the humidity requirement specified in the official minor-project guideline. (Project Guidelines, Section 2: Weather Monitoring Station)

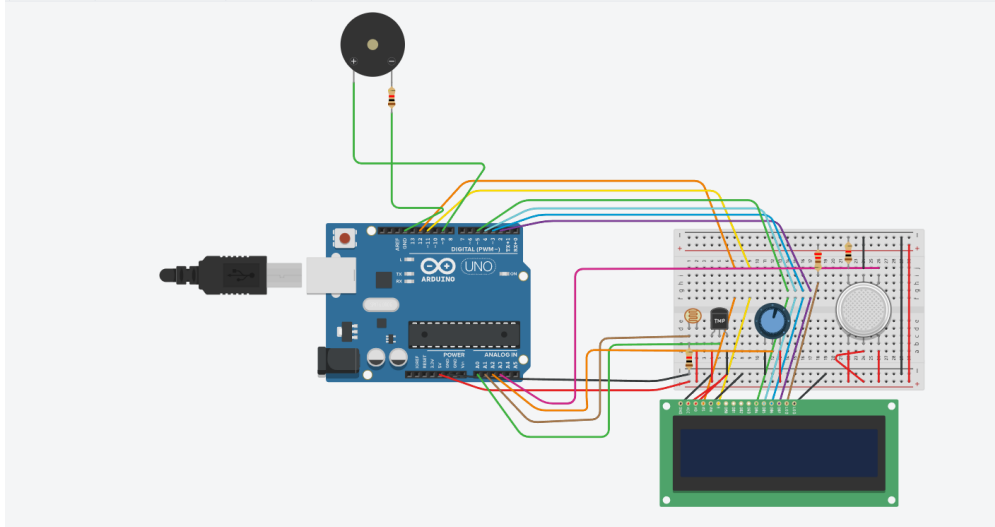


Figure 4. Schematic-level view of the integrated system.

Name	Quantity	Component
U3	1	Arduino Uno R3
2	1	Breadboard Small
U4	1	LCD 16 x 2
R2	1	220 Ω Resistor
U12	1	Temperature Sensor [TMP36]
R1 R3 R4	3	1 k Ω Resistor
Rpot1	1	250 G Ω Potentiometer
GAS1	1	Gas Sensor
R6	1	Photoresistor
PIEZ01	1	Piezo

Figure 5. Breadboard-style Tinkercad implementation with Arduino, sensors, LCD, and buzzer.

13. Challenges Faced

- Interfacing several sensors and output devices while maintaining common power and ground references.
- Organizing sensor information so that it can be displayed clearly on a limited 16x2 LCD.
- Choosing threshold values that produce meaningful alerts without excessive false alarms.
- Understanding how simulation wiring maps to a future physical breadboard implementation.
- Ensuring that the project scope remains aligned with the training-program requirements; in particular, humidity is not present in the current circuit and therefore remains an outstanding enhancement.

14. Conclusion

The project successfully demonstrates a small Arduino-based environmental monitoring and safety system in the Tinkercad simulation environment. The design integrates temperature, light, and gas-related sensing with a 16x2 LCD for local information display and a piezo buzzer for warning output. The architecture illustrates the fundamental embedded workflow of sensing, processing, decision making, and user notification.

The project is therefore a useful minor-project prototype for the IoT and Robotics training program. At the same time, the official guidelines specify that temperature and humidity measurement are mandatory for the Weather Monitoring Station project. The current design contains a TMP36 but no dedicated humidity sensor, so the report does not claim full compliance with that mandatory humidity requirement. Adding an appropriate humidity sensor and integrating it into the LCD display and testing plan would make the project more closely aligned with the specified requirements. (Project Guidelines, Section 2: Weather Monitoring Station)

15. Learning Outcomes

- Learned how an Arduino Uno can coordinate multiple analog sensors and output devices.
- Understood the role of ADC-based sensor reading and threshold comparison in embedded monitoring.
- Gained practical familiarity with 16x2 LCD interfacing and local user feedback.
- Learned how an audible actuator can be incorporated as a warning mechanism.
- Practiced reading and interpreting a schematic/BOM and translating it into a breadboard-style simulation.
- Gained experience with test planning and identifying limitations before physical deployment.

16. Future Scope

- Add a humidity sensor so that the mandatory temperature-and-humidity requirement is satisfied.
- Add Wi-Fi using ESP8266/ESP32 and publish data to ThingSpeak, Blynk, or another IoT platform, as encouraged by the guidelines. (Project Guidelines, Section 2: Weather Monitoring Station)
- Add data logging and historical trend graphs for temperature, humidity, light, and gas readings.
- Improve the alarm logic using calibrated thresholds, hysteresis, and separate warning/critical levels.
- Add a mobile/web dashboard and notifications for remote monitoring.
- Construct and test the design on physical hardware after validating all power, pin, and sensor requirements.

17. References and Project Basis

1. IoT & Robotics Training Program, "Minor & Major Project Guidelines," July Batch 2026. The submitted-report structure and testing expectations were based on the provided guideline document. (Project Guidelines, Sections 5-6: Deliverables and Report Requirements)
2. Project circuit, component list, and Tinkercad screenshots supplied with this submission.
3. Arduino Uno and common sensor interfacing concepts used in the Tinkercad simulation.